

Scenario 1: Multi-Company Sync with Zero-Data Entities

Connecting a QuickBooks company with 0 Classes and 0 Locations initially, then adding new Classes in QuickBooks and clicking **Refresh Schedule**.

- **Expected Result:**
 - Initial pull completes successfully, with **Classes (N:Q)** and **Locations (S:V)** remaining blank.
 - After refresh, newly added Classes are fetched and populated in **N:Q**.

A	B	C	D	E	F	G	H	I
Client ID	Client Name		Client ID	Account Code	Account Name	Account Type	Account Sub-Type	Classified
Advanced Sandbox Company US 866e	Advanced Sandbox Company US 866e		Advanced Sandbox Company US 866e		Accounting	Expense	Legal/Professional Fees	Expense
			Advanced Sandbox Company US 866e		Accounts Payable (A/P)	Accounts Payable	Accounts Payable	Liability
			Advanced Sandbox Company US 866e		Accounts Receivable (A/R)	Accounts Receivable	Accounts Receivable	Asset
			Advanced Sandbox Company US 866e		Advertising	Expense	Advertising/Promotion	Expense
			Advanced Sandbox Company US 866e		Arson/Debt of Revenue Payable	Other Current Liability	Global/Tax Payable	Liability
			Advanced Sandbox Company US 866e		Automobile	Expense	Auto	Expense
			Advanced Sandbox Company US 866e		Bank Charges	Expense	Bank Charges	Expense
			Advanced Sandbox Company US 866e		Billable Expense	Income	Service/Income	Revenue
			Advanced Sandbox Company US 866e		Board of Equalization Fee	Other Current Liability	Global/Tax Payable	Liability
			Advanced Sandbox Company US 866e		Bookkeeper	Expense	Legal/Professional Fees	Expense
			Advanced Sandbox Company US 866e		Building Repairs	Expense	Repair/Maintenance	Expense
			Advanced Sandbox Company US 866e		Checking	Bank	Checking	Asset
			Advanced Sandbox Company US 866e		Commissioners & Fees	Expense	Other/Miscellaneous	Expense
			Advanced Sandbox Company US 866e		Computer Repairs	Expense	Repair/Maintenance	Expense
			Advanced Sandbox Company US 866e		Cost of Goods Sold	Cost of Goods Sold	Supplies/Materials	Expense
			Advanced Sandbox Company US 866e		Cost of Labor	Expense	Other/Miscellaneous	Expense

FA

FinAccrual

STANDARD PLAN

N

Disconnect QuickBooks

Realm ID: 9341457814334726

QUICKBOOKS COMPANIES

+ Add Another Company

qb

Advanced Sandbox Company US 866e

Last Sync: Just now

ACTIVE

Showing 1 QuickBooks company

CURRENT PLAN

CURRENT PLAN

Standard

CONNECTED COMPANIES

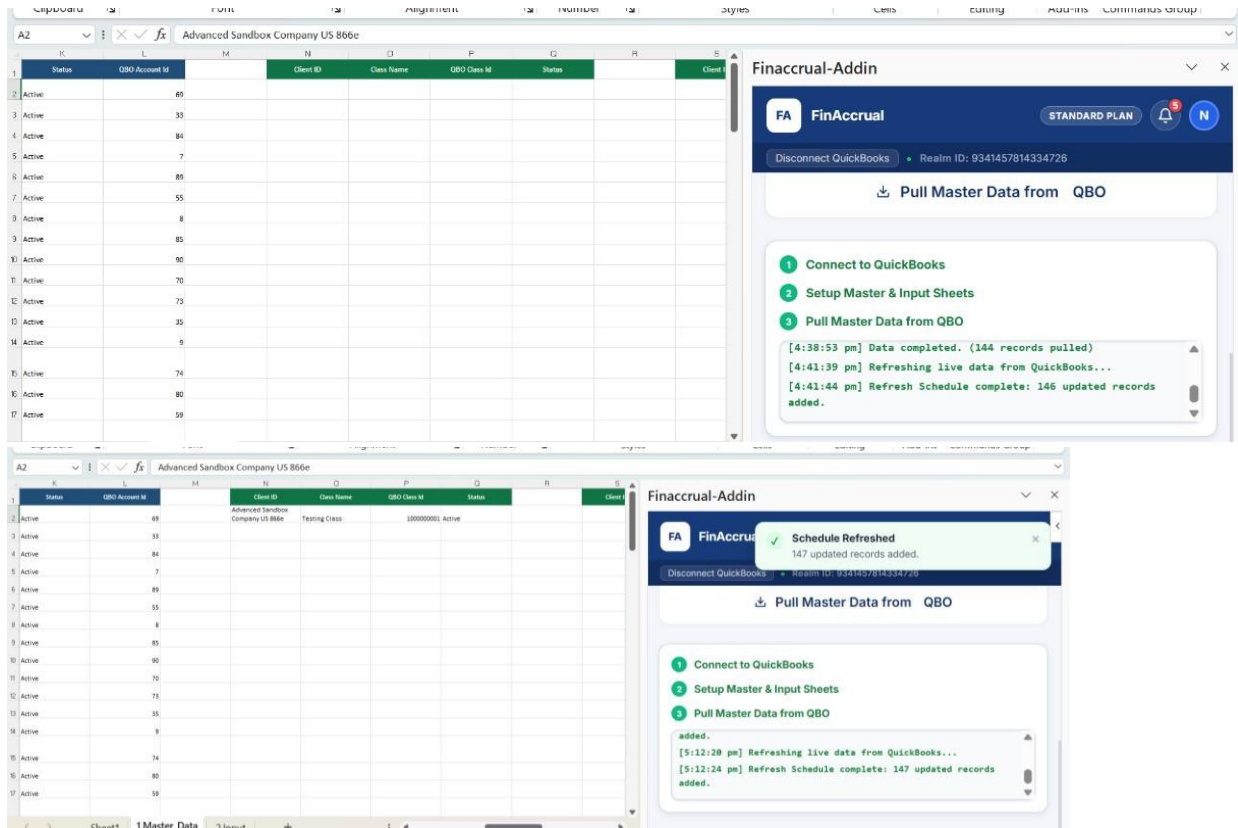
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REMAINING

2

Change Plan

Manage Companies



Technical Execution Flow

1)Independent OAuth & Session Scoping:

- Each company is identified by its own unique `realm_id` in the database QuickBooksToken.
- Company 1 and Company 2 possess **separate OAuth access and refresh tokens**. Their API calls are completely segregated.

2)Sequential Entity Querying:

In `QuickBooksService.pullMasterDataMultithreaded`, the backend go through active tokens:

Company 1 (realm_id_1): Calls Account, Class, Department, Customer, Vendor. Returns: accounts: [45], customers: [120], classes: [], locations: [], vendors: [98].

Company 2 (realm_id_2): Calls Account, Class, Department, Customer, Vendor.

Returns: accounts: [30], classes: [12], customers: [], locations: [], vendors: [78].

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3)Worksheet Rendering via Organization Blocks:

In ExcelService.writeMasterData, the frontend maps each entity into an orgGroupsMap:

Company 1 gets populated in rows 2–122. Columns **N:Q** (Classes) and **S:V** (Locations) remain clean and empty.

Company 2 gets populated in rows 124–154. Columns **N:Q** (Classes) are populated, while **X:AB** (Customers and Location) remain empty.

Handling these specific multi-company edge cases is essential for **data integrity, system resilience, and compliance**:

Scenario	What Happens If NOT Handled	Purpose of Handling in FinAccrual
Partial Session Expiry	A single revoked/expired company token crashes the entire synchronization job, blocking all other healthy companies from syncing.	Fault Isolation: Sync operations are independent per company. One failed connection logs the error and flags only that company as Disconnected, while healthy companies finish syncing smoothly.
Data Interleaving / Contamination	Chart of Accounts, Customers, or Vendors from <i>Company A</i> get merged into <i>Company B</i> , leading to inaccurate financial statements and severe accounting errors.	Multi-Entity Segregation: Every single entity is tagged (clientId, clientName) at the service layer to guarantee strict tenant isolation in memory and in Excel sheets.
Row Alignment Distortion	Because different companies have varying numbers of accounts, vendors, or classes, Excel rows would overwrite each other or skew data columns out of sync.	Worksheet Structure Integrity: Dynamic height computation (maxRows = Math.max(...)) ensures each organization's data block in Excel renders cleanly without overlapping or misaligned rows.
Plan Slot & Concurrency Limits	Users on lower pricing tiers could sync unlimited organizations, overloading system resources and violating subscription terms or QuickBooks API rate limits.	Business Rule & Tier Enforcement: Restricts multi-tenant concurrency to the user's licensed tier (e.g., slicing tokens according to plan allowances).

These protections are already built directly into our application codebase:

1. **Tagging / Isolation:** Implemented in Backend/src/modules/quickbooks/service.js:

Javascript:

```
const tag = (list) => list.map(i => ({ ...i, clientId: orgName, clientName: orgName }));
```

2. **Per-Company Error Isolation & Disconnection:** Implemented in Backend/src/modules/quickbooks/service.js with independent token iterations and status updates.

3. **Excel Row Alignment:** Implemented in Frontend/src/taskpane/services/excelService.js:

Javascript:

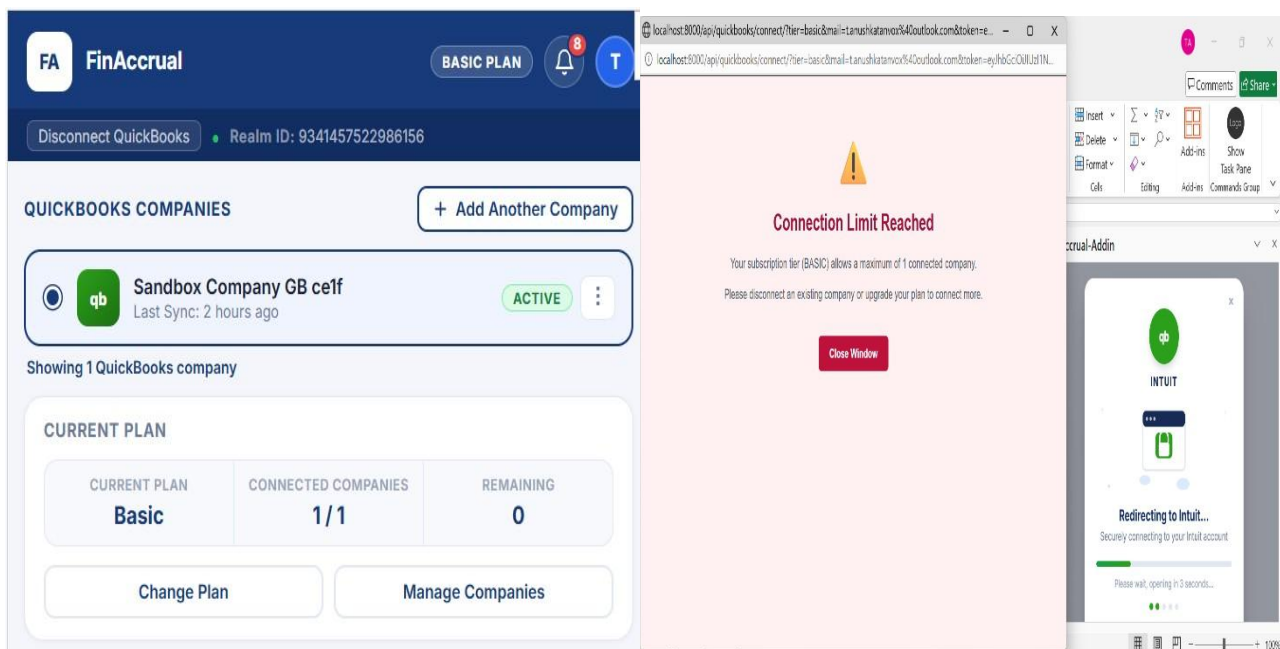
```
const maxRows = Math.max(1, accCount, classCount, locCount, entityCount); currentRow  
+= maxRows + 2;
```

4. **Plan / Slot Limits:** Implemented via QuickBooksService.getMaxConnections(tier).

- **Code Logic :** Error handling and business rules must always live in the core code **before** deployment. Since the code is already written, you do **not** have to rewrite it from scratch.
- **Server Deployment Phase:** When deploying the code to your server , what you need to ensure is:
 - Database schema is up to date (e.g., status, last_synced_at columns on QuickBooksToken).
 - Environment variables (Intuit Client ID/Secret, Redirect URIs, JWT Secrets) match your production/staging QuickBooks App.
 - Webhook/daemon handlers for background token refresh (if enabled).

Scenario 2: Plan Limit Enforcement & Upgrade Protection (Trial vs Standard vs Pro)

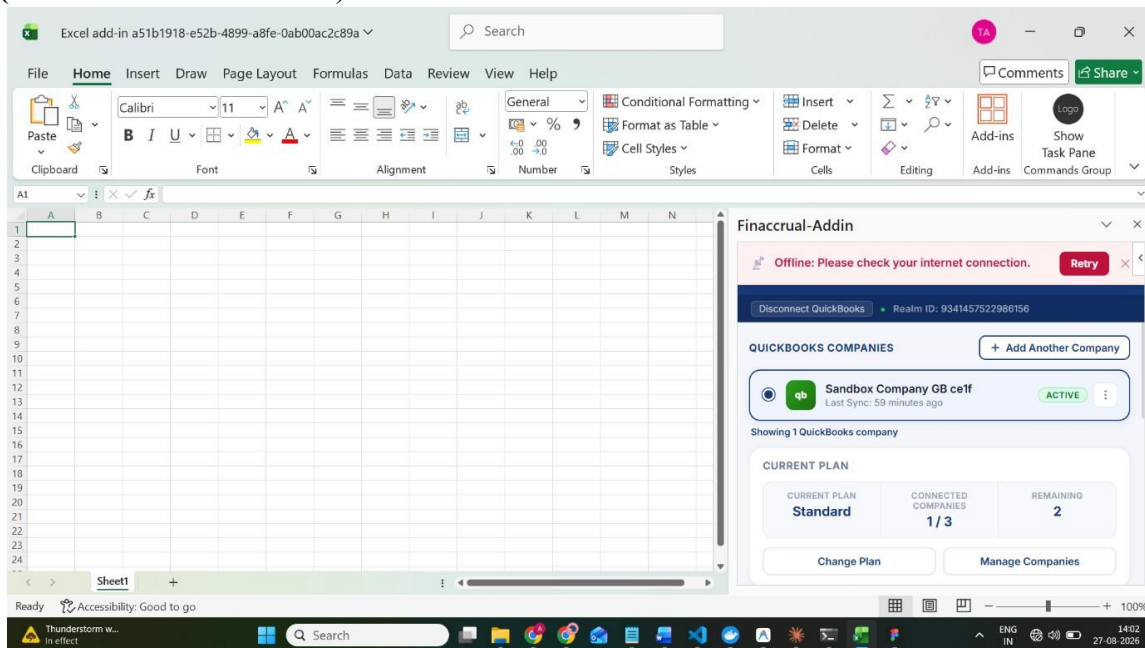
- A user on a **Trial / Basic plan** (allowed 1 company) tries to connect a 2nd QuickBooks company, or a **Standard user** (allowed 3 companies) tries to connect a 4th company.
- **Technical Execution Flow:**
- **Pre-Auth Check:** Before redirecting to QuickBooks, QuickBooksController.connectQuickbooks queries active tokens for the user's email: where: { status: { [Op.ne]: 'Disconnected' }, mail }.
- **Limit Interception:** If connectedCount >= maxAllowed, the backend blocks the OAuth redirection.
- **Callback Safeguard:** Even if a user attempts to alter query parameters midflight, quickbooksCallback validates the session-stored tier limit before saving any new token.
- **Expected Result:** The user is blocked from exceeding their quota and sees a branded error page: *"Connection Limit Reached: Your subscription tier allows a maximum of N connected companies. Please upgrade your plan."*



Scenario 3:Internet is Off / No Connectivity & Backend Issue / Server Down

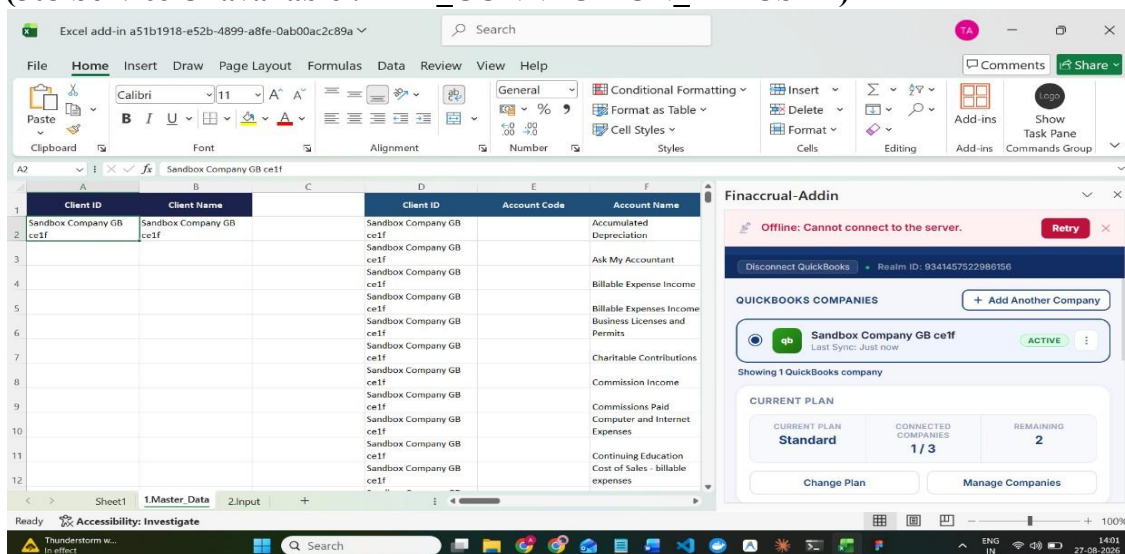
1)When Wi-Fi / Internet is Off , the user will see this notification and cannot access any other buttons

(503 Service Unavailable)



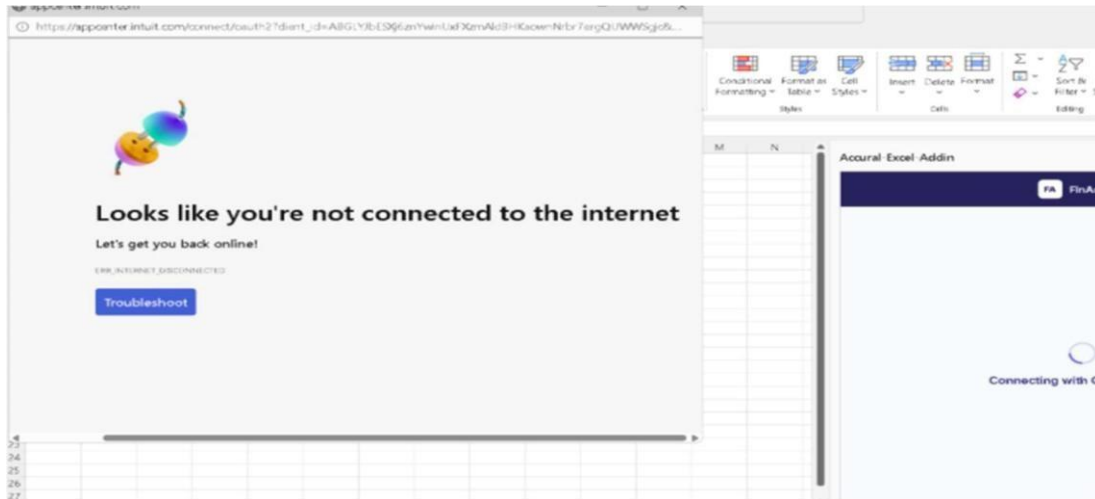
2)When Backend Server is Off (Wi-Fi connected, but backend unreachable) then user will see this notification (if user fetched data also it will display the same notification)

(503 Service Unavailable : ERR_CONNECTION_REFUSED)



3) authentication connection error

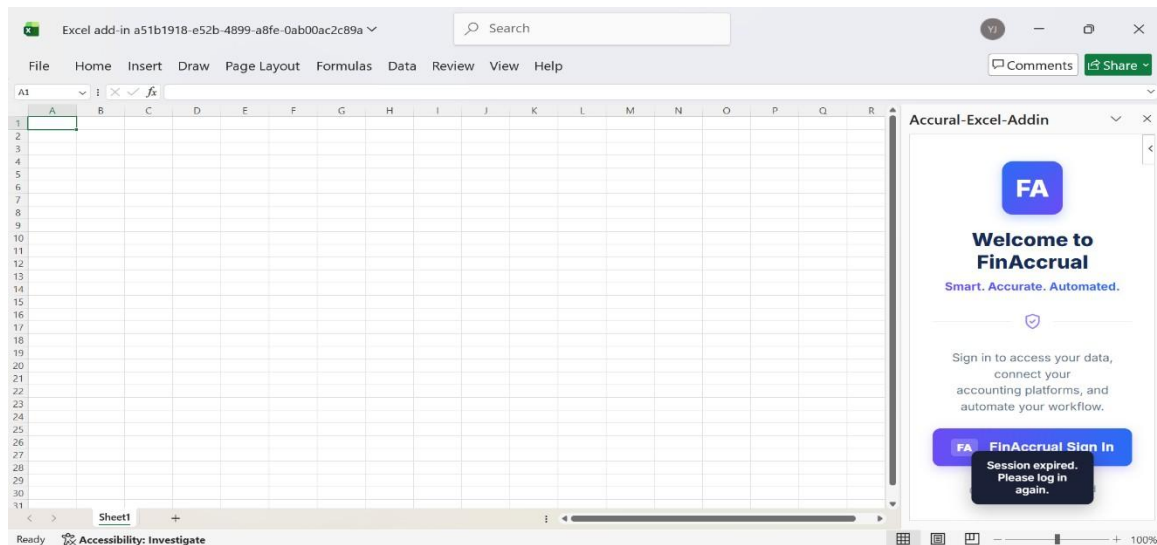
(503 Service Unavailable , Code: ERR_AUTH_CONNECTION_FAILED)



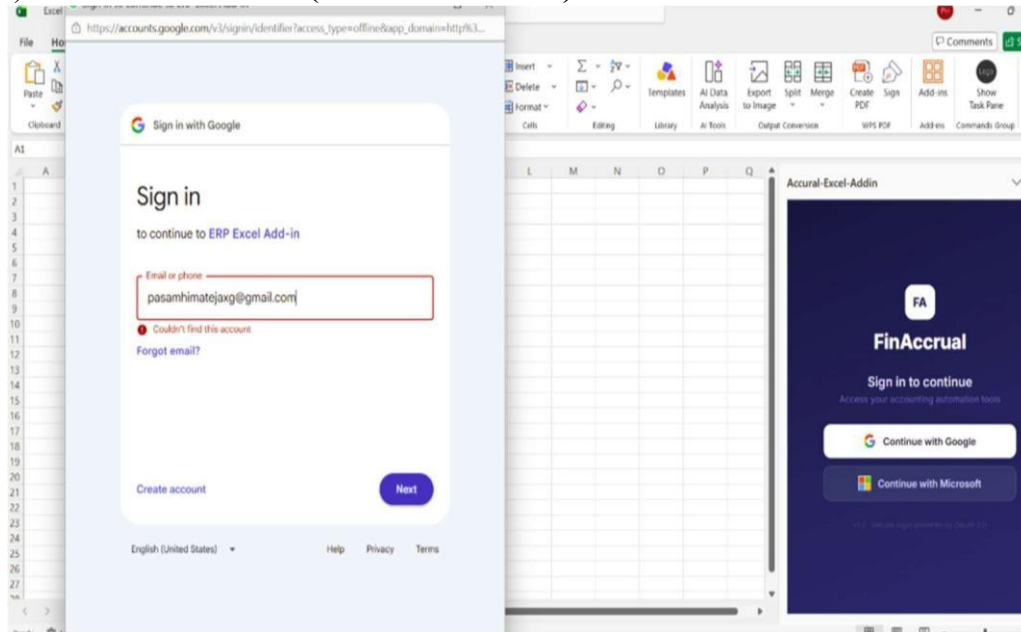
Scenario 4:

1)When the token expire , it will show the session expired please login again and it will redirect to the finaccrual dashboard . SessionExpiredError (JWT Expiry)

- **HTTP Status: 401 Unauthorized**
- **Code: ERR_SESSION_EXPIRED**



2) Unauthorized Error (401 Unauthorized)

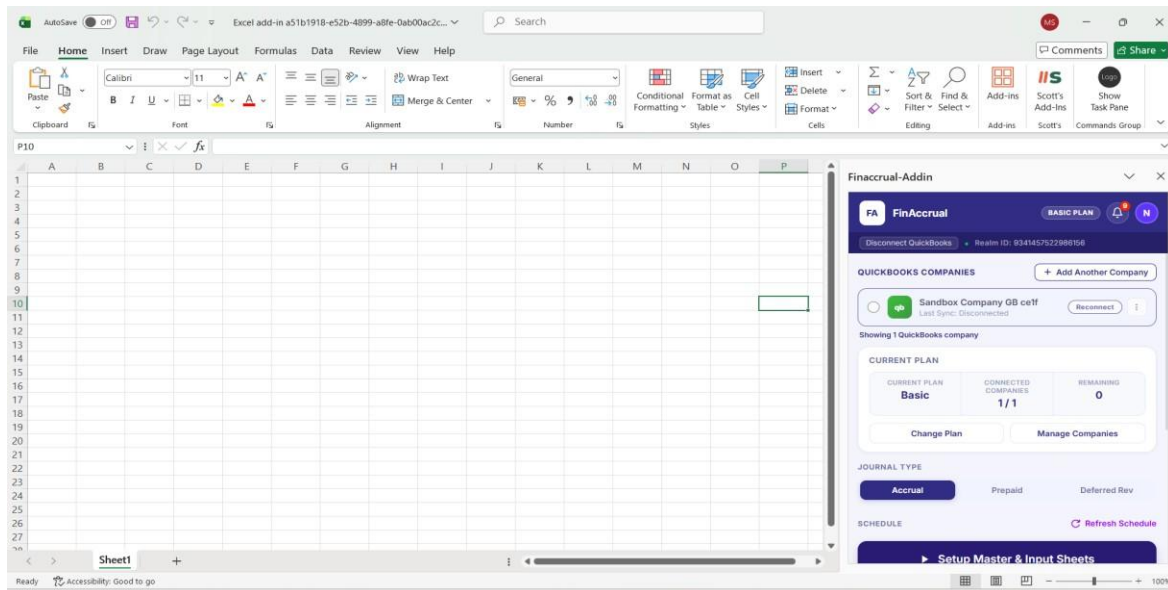


Scenario 5: Disconnect and reconnect workflow

1. When the "Reconnect" Button Shows Up

The "Reconnect" button is displayed in the **Company Management list** on your dashboard. When the dashboard loads, it fetches your companies from the backend. The code checks the status of each company:

- If the company is currently active: It shows "**ACTIVE**".
- If the company's status is 'Disconnected' (because you disconnected it previously or the token expired permanently): It displays a "Reconnect" button.

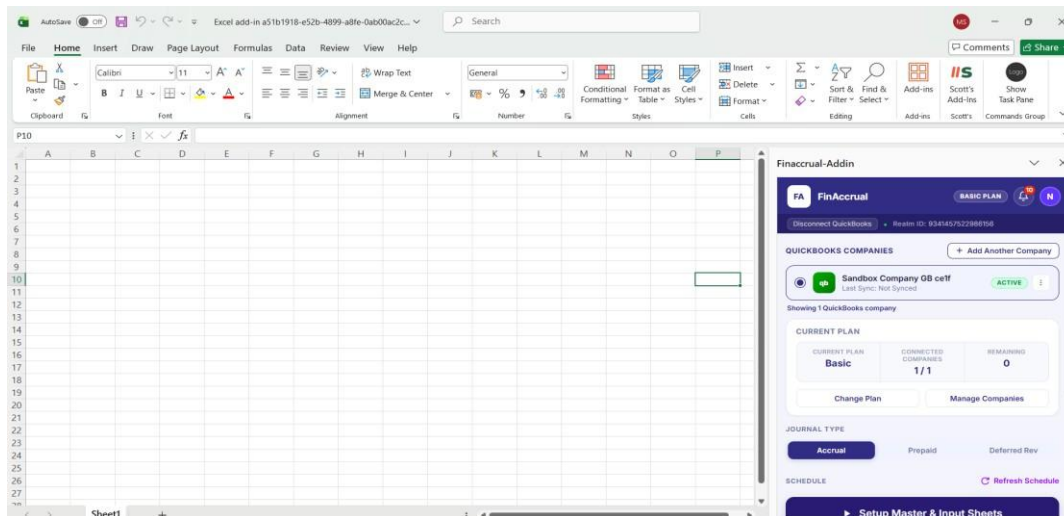


2. The Reconnect Flow

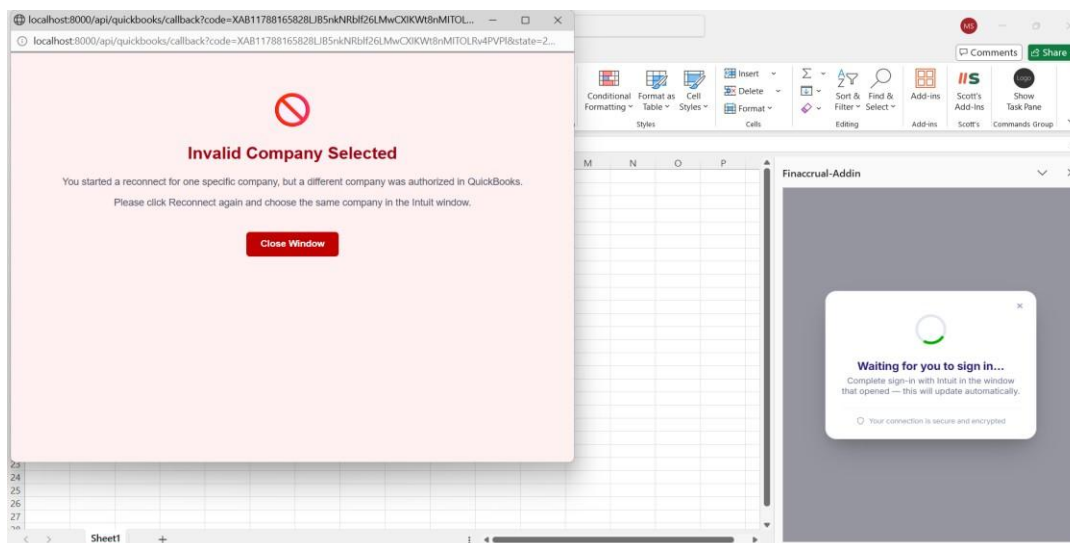
When you click the "Reconnect" button for a disconnected company:

1. **Targeted Authorization:** The app calls the `launchERPOAuth` function, but it specifically passes the `companyId` (as `reconnectId`) in the URL to the backend.
2. **Popup Opens:** The standard QuickBooks or Xero secure sign-in window opens.
3. **Backend Verification:** After you sign in, the backend receives the new authorization tokens. **Crucially**, it compares the company you just authorized against the `reconnectId`. This strict check ensures that you don't accidentally authorize *Company B* when you clicked reconnect on *Company A*.
4. **Restoration:** If it matches, the backend saves the new secure tokens and flips the company status back to Active.

Dashboard Update: The popup closes, the task pane refreshes, the "Reconnect" button turns back into "ACTIVE", and you can immediately resume syncing data to your Excel sheet.

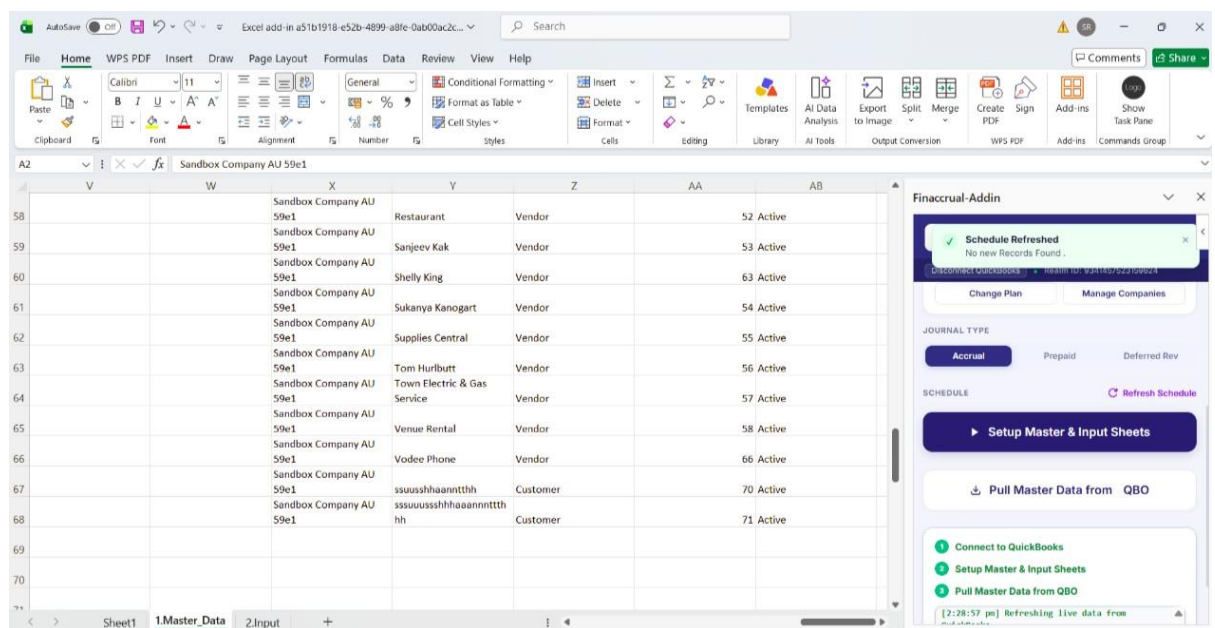


If you click **Reconnect** for a specific company (let's call it *Company A*) but then accidentally select a completely different company (*Company B*) inside the QuickBooks or Xero sign-in window, here is exactly what happens:

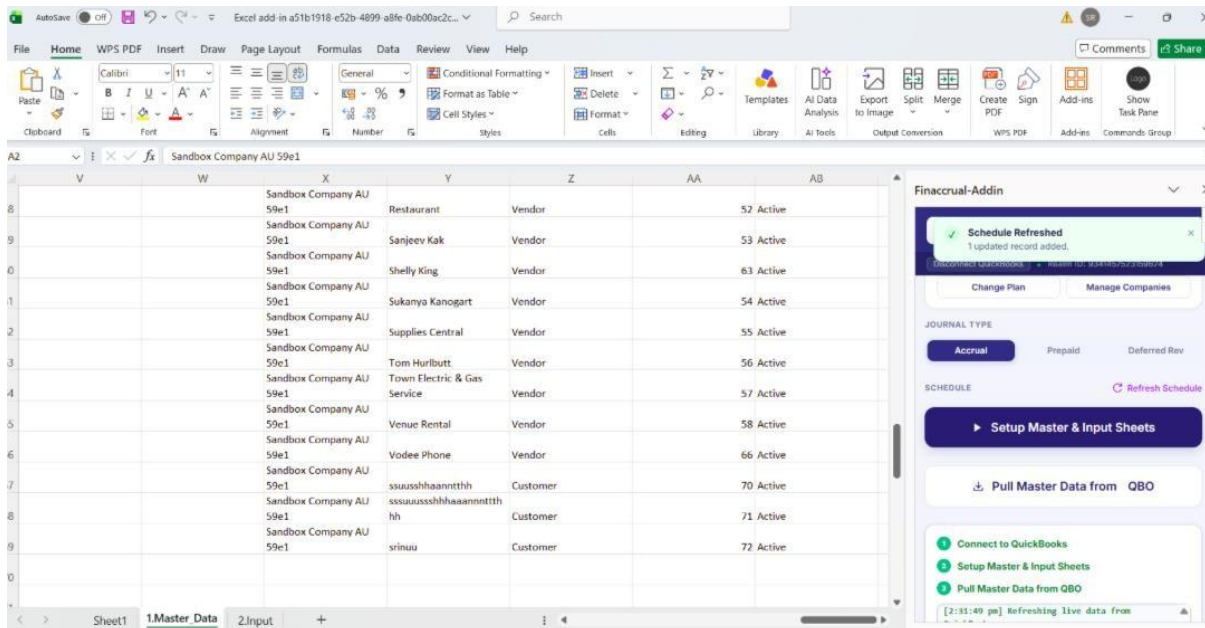


Scenario 6: Refresh functionality

- Click **Refresh** → front-end calls `/api/pull-master-data?mode=incremental`.
- Backend queries QuickBooks/Xero for records with `MetaData.LastUpdatedTime >= last_synced_at`.
- If no records are returned, `last_synced_at` is still updated and the UI shows **“Schedule Refreshed – No new Records Found.”**



- If records are returned, they are appended to the Excel sheet and a toast **“N updated record(s) added.”** is shown.



	V	W	X	Y	Z	AA	AB
8			Sandbox Company AU 59e1	Restaurant	Vendor		52 Active
9			Sandbox Company AU 59e1	Sanjeev Kak	Vendor		53 Active
0			Sandbox Company AU 59e1	Shelly King	Vendor		63 Active
1			Sandbox Company AU 59e1	Sukanya Kanogart	Vendor		54 Active
2			Sandbox Company AU 59e1	Supplies Central	Vendor		55 Active
3			Sandbox Company AU 59e1	Tom Hurlbutt	Vendor		56 Active
4			Sandbox Company AU 59e1	Town Electric & Gas Service	Vendor		57 Active
5			Sandbox Company AU 59e1	Venue Rental	Vendor		58 Active
6			Sandbox Company AU 59e1	Vodee Phone	Vendor		66 Active
7			Sandbox Company AU 59e1	ssuussshhaanntthh	Customer		70 Active
8			Sandbox Company AU 59e1	hh	Customer		71 Active
9			Sandbox Company AU 59e1	srinuu	Customer		72 Active